

Verified compiler for a language with control effects

(Weryfikacja kompilatora dla języka programowania z efektami sterowania)

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Abstract

We present a type system for a language with labeled delimited control operators. It utilizes polymorphic types and effects to allow expressions parametrized by effects. Our goal is to formally prove the correctness of such a language. Formalisation work was done in Agda programming language.

Tematem tej pracy jest stworzenie i zweryfikowanie języka z operatorami sterowania dla kontynuacji ograniczonych. Język ten używa polimorficznych typów aby pozwolić na wyrażenie obliczeń generycznych wobec pewnych efektów. Celem pracy jest zweryfikowanie poprawności definicji takiego języka. Formalizacja została przeprowadzona w języku programowania Agda.

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Chapter 1

Introduction

Control operators have been present in languages for a long time. One notable example is the SCHEME programming language which originated in the 70s[1] and provides a *call/cc* operator. Such operators enable non-local jumps, which might be familiar to users of imperative languages with *goto* operator. But those operators unlike *goto* are handled by capturing continuation, which is stored as first-class value, able to be passed, called or dropped. Operator *call/cc* when called, catches whole rest of program in the continuation. When this continuation is called, it aborts rest of the program and replaces it with this continuation. Thus in example program (*call/cc* (*k* . *k* (*k* 1))) only inner *k* would be called.

Operators for delimited control introduced by Danvy, and Filinski[2] and independently by Felleisen[3] allow to control continuations more precisely. They usually come in pairs: for example, the operator *reset* delimits continuations caught by *shift*. This allows localized use of control effects, as the scope of the captured continuation much more visible, while still allowing non-local execution, which enables interesting computational effects such as emulating non-determinism, exceptions and failure. Usually such calculi jump from *shift* to the closest enclosing *reset*. Reduction rule looks like this

$(\text{reset } E[(\text{shift } k \text{ expr})]) \Rightarrow (\text{reset } ((k . \text{expr}) (v . (\text{reset } E[v]))))$ therefore, for example $(\text{reset } (+ 1 (\text{shift } k (k (k 0)))))$ would call *k* twice and result in value 2.

Danvy and Filinski[4] also describe similar operators *shift*₀ and *reset*₀. Their reduction does not introduce outermost *reset*₀, therefore subsequent calls to *shift*₀ remove innermost *reset*₀ one by one. Their example showcases this behaviour *reset*₀(3 + *reset*₀(4 * *shift*₀ *k* in *shift*₀ *c* in *E*)). The continuation *k* would be bound to *x* . 4 * *x* and *c* to *x* . 3 + *x*. In addition Dybvig et al.[5] show the formalisation of delimited control with multiple prompts where labels are used to match the corresponding operators.

The aim of this thesis is to formalise a typed version of such a calculus with poly-

morphic types (similar to Piróg, Polesiuk and Sieczkowski[6]), but for labeled delimited continuations (like Kazuki Ikemori, Youyou Cong, and Hidehiko Masuhara[7]). Compared to Ikemori et al., the main difference in our work is the use of first-class labels for delimited continuations, similarly to how Xie et al.[8] used them for algebraic effects. Balik and Polesiuk presented such a language[9] which currently serves as the core of Fram programming language. Formalising such a calculus could serve as a step in to demonstrate soundness of Fram’s implementation.

The soundness of the presented language is proved using the standard technique of progress and preservation[10][11]. Formalisation work was done in Agda 2.8. The code archive also features nix derivation describing the exact environment to build both package and this paper to ensure reproducibility.

Chapter 2

Surface language

2.1 Syntax

We start with presenting the syntax of the surface language. We use kinds to distinguish between types and effects.

```
data Kind : Set where
  T : Kind
  E : Kind
```

Types and effects are defined as mutually recursive. A type is either a type variable `ttv`, an arrow, a `forallt` or a label `L eff at A / E`. We represent variables and type variables with de Bruijn indices. Effects datatype stores list of effects (represented as types), it's used in arrow and forall constructors of type to mark effects of computation underneath. It will be also used in typing judgements to mark effects available in computation. Constructor `L` represents type of label expressions. It just stores an effect `eff` represented by a label and then type and effect of delimited computation labelled by this label.

```
module Types where
  Id : Set
  Id = ℕ
  data Type : Set
  Effects : Set
  data Type where
    ttv : Id → Type
    -->_ : Type → Effects → Type → Type
    forallt : Kind → Type → Effects → Type
    Lat_/_ : Type → Type → Effects → Type
  Effects = List Type
  open Types
```

Constructors of expressions such as `var`, `lam`, `app`, `tlam`, and `tapp` behave as usual. `var` is a de Bruijn indexed variable, `lam` is a lambda abstraction, `app` is a function application, `tlam` is a type abstraction, storing kind of abstracted type and `tapp` is type application. Other constructors are responsible for continuations and labels. Constructor `new` binds labels used by `shift0` and `reset0` expressions to pair them up. Constructor `shift0` stores the label and expression (parametrized

by continuation) that replaces whole delimited computation. Last constructor of expressions is `reset0 e (v . en) e'` which has 3 arguments, last one `e'` is label that is used to match up `reset0s` with `shift0s`. First argument `e` is delimited computation where `shift0` can be used. So when `shift0 k.es` is being evaluated, it aborts enclosing up to corresponding `reset0`, replacing it with `es` is being evaluated with `k` bound to continuation representing evaluation context between `shift0` and its `reset0`. Second argument `x. en` is used when no corresponding shift aborts during evaluation of `e`. If `e` evaluates to `v` - then whole `reset0` reduces to `en` where `x` is bound to `v`.

```
module Expr where
  data Expr : Set where
    var : N → Expr
    lam : Expr → Expr
    app : Expr → Expr → Expr
    tlam : Kind → Expr → Expr
    tapp : Expr → Type → Expr
    new : Expr → Expr
    shift0 : Expr → Expr → Expr
    reset0 : Expr → Expr → Expr → Expr
  open Expr
```

2.2 Typing judgements

Typing contexts are represented as lists of types, and contexts for type variables are represented as lists of kinds. Type or kind judgements for membership have Peano numbers structure.

```
module Typing where
  infixl 5 _>_
  data Context : Set where
    ∅ : Context
    _>_ : Context → Type → Context

  data TContext : Set where
    ∅ : TContext
    _>_ : TContext → Kind → TContext
```

Expressions are extrinsically typed, thus typing judgements are represented separately. Typing rules for `var`, `lam`, `app`, `tlam`, and `tapp` are defined mostly as usual. The noticeable difference is that value expressions are generic over effects they execute.

```
data _>_>_>_>_>_ : TContext → Context → Expr → Type → Effects → Set where
  ⊢-var : ∀ {Γ Δ x A E}
    → Γ ⊇ x ∷ A
    -----
    → Δ , Γ ⊢ var x ∷ A / E

  ⊢-lam : ∀ {Γ Δ e A B E F}
    → Δ , (Γ , A) ⊢ e ∷ B / E
    -----
    → Δ , Γ ⊢ lam e ∷ A - E > B / F

  ⊢-weak : ∀ {Γ Δ e A A' E E'}
    → Δ ⊢ A <ₜ A'
    → Δ ⊢ E <ₛ E'
    → Δ , Γ ⊢ e ∷ A / E
    -----
    → Δ , Γ ⊢ e ∷ A' / E'
```

```

 $\vdash_{\text{app}} : \forall \{ \Gamma \Delta e_1 e_2 A B E \}$ 
 $\rightarrow \Delta, \Gamma \vdash e_1 : A \multimap E \multimap B / E$ 
 $\rightarrow \Delta, \Gamma \vdash e_2 : A / E$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{app } e_1 e_2 : B / E$ 

 $\vdash_{\text{forall}} : \forall \{ \Gamma \Delta e k A E F \}$ 
 $\rightarrow (\Delta, k), \Gamma \vdash e : \text{TypeSubst.bump } A / \text{TypeSubst.bump}' E$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{tlam } k e : \text{foralllt } k A E / F$ 

 $\vdash_{\text{tapp}} : \forall \{ \Gamma \Delta e k A T E \}$ 
 $\rightarrow \Delta \vdash T : \text{ste}$ 
 $\rightarrow \Delta, \Gamma \vdash e : \text{foralllt } k A E / E$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{tapp } e T : A \text{TypeSubst.}[ T ] / (E \text{TypeSubst.effs}[t T ])$ 

```

The **new** construct introduces a new type variable and a variable that represent respectively an effect and a label. The label type stores effect bound by **new** (here **ttv zero**) and **A1 / E2** representing a type and an effect of delimited computation.

```

 $\vdash_{\text{new}} : \forall \{ \Gamma \Delta e A A_1 E E_1 \}$ 
 $\rightarrow \Delta \vdash A_1 : \text{st}$ 
 $\rightarrow \Delta \vdash E_1 : \text{effs}$ 
 $\rightarrow (\Delta, \text{Kind.E}), (\Gamma, (L \text{ttv zero at } A_1 / E_1)) \vdash e : \text{TypeSubst.bump } A / \text{TypeSubst.bump}' E$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{new } e : A / E$ 

```

Constructor **shift₀ e' (k . e)** uses only one effect **E** represented by label **e'**. For it to be a properly typed expression inside shift, the constructor binds extra variable **k**, which is where continuation will be plugged. Therefore the type of this expression **e** should be the same type and effect as stored in label type. Its typing context should be expanded to account for continuation, which type should be an arrow from **shift₀** type to type and effect of whole delimited computation. Since during evaluation **reset₀** will be removed, effect **E** it has introduced will not be present in direct subexpressions of **shift₀**.

```

 $\vdash_{\text{shift}_0} : \forall \{ \Gamma \Delta e e' A A' E E' \}$ 
 $\rightarrow \Delta \vdash E : \text{se}$ 
 $\rightarrow \Delta, \Gamma \vdash e' : (L E \text{at } A' / E') / \text{nil}$ 
 $\rightarrow \Delta, (\Gamma, A \multimap E' \multimap A') \vdash e : A' / E'$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{shift}_0 e' e : A / (E :: \text{nil})$ 

```

The **reset₀ e (v . en) e'** constructor has three parameters. The first one is expression **e** that will have access to the effect, so its list of effects is expanded. The second one is continuation **v . en** that will handle the value **v** returned from the first argument **e**. And third is the label **e'**, whose effect is the same one as one introduced in **e**. Label type also stores the type and effects of the whole delimited computation.

```

 $\vdash_{\text{reset}_0} : \forall \{ \Gamma \Delta e e' en A A' E E' \}$ 
 $\rightarrow \Delta \vdash E : \text{se}$ 
 $\rightarrow \Delta, \Gamma \vdash e : A / (E :: E')$ 
 $\rightarrow \Delta, (\Gamma, A) \vdash en : A' / E'$ 
 $\rightarrow \Delta, \Gamma \vdash e' : (L E \text{at } A' / E') / \text{nil}$ 
-----
 $\rightarrow \Delta, \Gamma \vdash \text{reset}_0 e en e' : A' / E'$ 

```

Now that the surface language and its typing rules have been defined for it, the next chapter will consider how it could be evaluated.

Chapter 3

Runtime language

3.1 Runtime language

Grammar of terms we defined previously doesn't have any expression that would have type of label, and shift and reset require the same labels. That means that to define reduction relation we would need to either go under **new** binders or introduce label expressions. We decided to define another runtime expression language expanded by the notion of label values.

When **new** expression is evaluated then all occurrences of variables bound by it would have it replaced with newly allocated label value. That means evaluation would need to keep a state for the allocator.

To ensure type safety we will expand syntax of types with allocated effects, and add new effect context to typing judgments, that maps allocations to type and effect of delimiter.

```
module Types_ where
  Id : Set
  Id = ℕ
  Label = ℕ
  data Type : Set
  Effects : Set
  data Type where
    ttv : Id → Type
    _->_ : Type → Effects → Type → Type
    forallt : Kind → Type → Effects → Type
    L_at_/_ : Type → Type → Effects → Type
    Effect : Label → Type
  Effects = List Type
open Types_
```

Most of the constructors in **RExpr** are the same as in **Expr**. Labels runtime values are represented by natural numbers.

```
module RuntimeExpr where
  data RExpr : Set where --runtime version
    var : ℕ → RExpr
    lam : RExpr → RExpr
    app : RExpr → RExpr → RExpr
    tlam : Kind → RExpr → RExpr
    tapp : RExpr → Type → RExpr
    new : RExpr → RExpr
```

```

shift0 : RExpr → RExpr → RExpr
reset0 : RExpr → RExpr → RExpr → RExpr

```

And here we have a separate term for the label, it just stores the allocated label identifier.

```

label : Label → RExpr

```

Since the grammar of types has changed, we need to redefine contexts. That means almost all of the typing judgements need to be redefined. Here we also introduce a new context for effects that maps label values to type and effect of delimiter. Such technique of store typing is described by Pierce[12] where to ensure type safety of allocations, typing rules take maps from locations to types. It's represented by a finite vector of types and effects. Since they are only used during evaluation, the types and effects have no free variables.

```

module Typing where
open RuntimeExpr
infixl 5 _,-
data Context : Set where
  ∅ : Context
  _,- : Context → Type → Context

data TContext : Set where
  ∅ : TContext
  _,- : TContext → Kind → TContext

push : Kind → TContext → TContext
push k Δ = Δ , k
EContext = Σ[ n ∈ ℕ ] Data.Vec.Vec (Type × Effects) n
pattern ∅' = ∅ ,, Data.Vec.[]

```

Most of the typing judgements are working as before. We omit ones that show up in surface language, as they just pass extra context for effects throughout.

```

private
variable
  Γ : Context
  Δ : TContext
  Θ : EContext
  e e1 e2 : RExpr
  A B : Type
  E E' F : Effects
  k : Kind
infix 4 _;-;_+;_&_/_-
data _;-;_+;_&_/_- : TContext → EContext → Context → RExpr → Type → Effects → Set where

```

Since labels are represented by natural numbers, ensuring type of label is just a lookup in appropriate context.

```

⊢label : ∀ {n }
  → Θ ⊢l n ∶ A / E
  -----
  → Δ ; Θ ; Γ ⊢ label n ∶ (L (Effect n) at A / E) / F

```

3.2 Runtime embedding

We defined embedding of the original language in the current one, and proof that such embedding preserves typing judgements. Most of the proof is omitted as it goes through almost all judgement types. We present only the type of main proof.

```
runtime-types : ∀ {Δ Γ e T E}
  → Δ Types.Typing., Γ ⊢ e : T / E → (runtimeΔ Δ ; ⋈' ; runtimeΓ Γ ⊢ (runtime e) : rt-t T / rt-t' E)
```


Chapter 4

Semantics

In this chapter we will define reduction relation and show that it is sound. # Values First, we need to define values. Only abstractions, type abstractions and labels are considered values. Since values themselves don't perform any effects, they have `nil` effect. But rules for all of them have built-in weakening. We can use that to generalize their type and perform substitution where any effect is expected.

```
data Value : REpr → Set where
  vLam : ∀ { e } → Value (lam e)
  vLam : ∀ { k e } → Value (tlam k e)
  vLab : ∀ { n } → Value (label n)
gvalue : ∀ {Δ Θ Γ T E e} → (Value e) → (Δ ; Θ ; Γ ⊢ e :: T / E) → ∀ {F} → (Δ ; Θ ; Γ ⊢ e :: T / F)
gvalue vLam (hlam t) = hlam t
gvalue vLam (hforall t) = hforall t
gvalue vLab (hlabel x) = hlabel x
gvalue {Δ} v (hweak x x1 x2) {F} = (hweak x ( <?e-refl {Δ} {F} ) (gvalue v x2))
```

4.1 Frames

What is called frame here would usually be referred to as evaluation context, but here the name is already taken by typing context. Here frame represents parts between `resetΘ`, or `resetΘ` and `shiftΘ`. Frame type is parametrized by $\Theta \Gamma$ - typing context outside of frame, T type of the hole. It's also indexed by Effects and Type of whole frame, Effects of the hole, typing context of the hole. Frames here are intrinsically typed, thus they also store type judgements of subexpressions. They are defined in such a way to reduce repetition, as otherwise we would need to introduce typing judgements for frames, and for every operation such as plugging or composition we would need to define it and then prove type preservation.

```
data Frame (Θ : EContext) (Γ : Context) (T : Type) : Effects → Type → Effects → Set where
  fempty : ∀ {Eff}
    -----
    → Frame Θ Γ T Eff T Eff
  fapp1 : ∀ {A B Eff E} → Frame Θ Γ T Eff (A - Eff > B) E
    → (e : REpr) → { Δ ; Θ ; Γ ⊢ e :: A / Eff }
    -----
    → Frame Θ Γ T Eff B E
  fapp2 : ∀ {A B Eff E} → (e : REpr) → { v : Value e }
    → { Δ ; Θ ; Γ ⊢ e :: (A - Eff > B) / Eff }
```

```

-----
→ Frame Θ Γ T Eff A E → Frame Θ Γ T Eff B E
freset-label : ∀ {A E A' Eff C}
→ (e en : RExpr)
→ Ø ; Ø ⊢ C %e
→ Ø ; Ø ; Γ ⊢ e % A' / (C :: Eff)
→ Ø ; Ø ; (Γ , A') ⊢ en % A / Eff
→ Frame Θ Γ T nil (L C at A / Eff) E
-----
→ Frame Θ Γ T Eff A E
fshift-label : ∀ {A E A' E' C}
→ (e : RExpr)
→ Ø ; Ø ⊢ C %e
→ Ø ; Ø ; (Γ , A - E' > A' ) ⊢ e % A' / E'
→ Frame Θ Γ T nil (L C at A' / E') E
-----
→ Frame Θ Γ T (C :: nil) A E

```

Definition and types for frame plugging and composition:

```

plug : ∀ {Θ Γ T Eff A E}
→ Frame Θ Γ T Eff A E
→ (e : RExpr) → Ø ; Ø ; Γ ⊢ e % T / E
→ Σ[ res ∈ RExpr ] (Ø ; Ø ; Γ ⊢ res % A / Eff)
_of_ : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ Frame Θ Γ B Eff A Eff'
→ Frame Θ Γ C Eff' B Eff''
→ Frame Θ Γ C Eff A Eff''

```

We can prove how plugging and composition relate. Plugging expression into one frame and another results in same value and type as plugging expression into composition of two frames.

```

of-lemma : ∀ {Γ Θ Eff Eff' Eff'' A B C}
→ (f1 : Frame Θ Γ B Eff A Eff' )
→ (f2 : Frame Θ Γ C Eff' B Eff'' )
→ (e : RExpr) → (t : Ø ; Ø ; Γ ⊢ e % C / Eff'')
→ plug ( f1 of f2 ) e t
≡ ((λ x → plug f1 (Data.Product.proj₁ x) (Data.Product.proj₂ x))(plug f2 e t))

```

Lifting frames into arbitrary context preserves types, same as with expressions.

```

↑f : forall { Θ A B Eff Eff' Γ' Γ }
→ Frame Θ Γ A Eff B Eff'
→ Frame Θ (Γ' # Γ) A Eff B Eff'

```

Metaframe stores the whole evaluation context, it's split into frames separated by resets. Type parameters and indices work the same as in frame. Unlike in frame, metaframe now stores resets, so lists of effects inside and outside of frame may differ. That means their difference represents a list of effects handled by the frame. This observation can be used to prove that for well typed expression (in empty typing context, and with condition that same labels have the same type) that decomposes into metaframe and `shift0` expression, and metaframe should handle effect of the `shift0`. Also this metaframe decomposes into two metaframes separated by `reset0` which has same label as mentioned `shift0`.

```

data Metaframe (Θ : EContext) (Γ : Context) (T : Type) (Eff : Effects)
: Type → Effects → Set where
mempty : Metaframe Θ Γ T Eff T Eff
mreset : ∀ { A B C Eff' }
→ (l : Label)
→ Ø ; Ø ⊢ C %e
→ Ø ; Ø ; Γ ⊢ label l % (L C at B / Eff) / nil

```

```

→ (e : REpr) → (∅ ; ∅ ; Γ , A ⊢ e :: B / Eff)
→ Metaframe ∅ Γ T (C :: Eff) A Eff'
-----
→ Metaframe ∅ Γ T Eff B Eff'
mframe : ∀ {A Eff' B Eff''}
→ Frame ∅      Γ A Eff B Eff'
→ Metaframe ∅ Γ T Eff' A Eff''
-----
→ Metaframe ∅ Γ T Eff B Eff'

```

Metaframes, same as frames, can be lifted into arbitrary contexts, plugged and composed. They can also be composed with simple frames.

```

tm : forall {∅ A B Eff Eff' Γ' Γ}
→ Metaframe ∅ Γ      A Eff B Eff'
→ Metaframe ∅ (Γ' + Γ) A Eff B Eff'
mplug : ∀ {∅ Γ T Eff A E}
→ Metaframe ∅ Γ T Eff A E
→ (e : REpr) → ∅ ; ∅ ; Γ ⊢ e :: T / E
→ Σ[ res ∈ REpr ] (∅ ; ∅ ; Γ ⊢ res :: A / Eff)
_om_ : ∀ {Γ ∅ Eff Eff' Eff'' A B C}
→ Metaframe ∅ Γ B Eff A Eff'
→ Metaframe ∅ Γ C Eff' B Eff''
→ Metaframe ∅ Γ C Eff A Eff''
_fom_ : ∀ {Γ ∅ Eff Eff' Eff'' A B C}
→ Frame ∅ Γ B Eff A Eff'
→ Metaframe ∅ Γ C Eff' B Eff''
→ Metaframe ∅ Γ C Eff A Eff''

```

4.2 Reduction

Since labels need to be allocated, the reduction relation is defined in terms of expression and state. State itself is just the next label to be allocated. As frames are intrinsically typed, we need to provide judgment representing well-typedness of expressions.

```

infix 2 _→_
State = EContext
data _→_ : REpr × State → REpr × State → Set where
{-
→new : ∀ {e Δ ∅ E T A1 E1}
→ {te : (Δ , Kind.E) ; ∅ ; (∅ , L ttv zero at A1 / E1) ⊢ e :: E / T}
→ new e , ' ∅ → REprSubstTyped._[_] e (Label (∅ .proj1)) {te = te} {te1 = {!!!}} .proj1 , ' ∅ --(Data.Vec._++_ ∅ ( Data.Vec
-})
β-lam-app : ∀ {e V ∅ Γ A B E}
→ {te : ∅ ; ∅ ; (Γ , A) ⊢ e :: B / E}
→ {tv : ∅ ; ∅ ; Γ ⊢ V :: A / E}
→ Value (lam e)
→ (v : Value V)
→ app (lam e) v , ' ∅ → (proj1 (REprSubstTyped._[_] e v {te = te} {te1 = (gvalue v tv)})) , ' ∅

β-tlam-tapp : ∀ {k e T ∅}
→ Value (tlam k e)
→ tapp (tlam k e) T , ' ∅ → REprSubst.e[t T] , ' ∅

reset0-vl : ∀ {V e' en ∅ Γ A B E}
→ (v : Value V)
→ {tv : ∅ ; ∅ ; Γ ⊢ V :: A / E}
→ {ten : ∅ ; ∅ ; (Γ , A) ⊢ en :: B / E}
→ reset0 v en e' , ' ∅ → (REprSubstTyped._[_] en v {te = ten} {te1 = (gvalue v tv)} .proj1) , ' ∅

{-
reset0-k : ∀ {es en e' e s n ∅ A T Eff Eff' B A' E' ls lr}
→ {f : Metaframe ∅ ∅ T Eff A Eff' }
→ ∀ {cont-type}
→ Value (e')
--shift
→ {ts : ∅ ; ∅ ; ∅ ⊢ (shift0 e' es) :: T / Eff'}
→ {tes : ∅ ; ∅ ; (∅ , T - Eff' > B) ⊢ es :: B / Eff' }

```

```

→ {tts :  $\emptyset$  ;  $\emptyset$  ;  $\emptyset \vdash e' \S (L \text{ ttv } ls \text{ at } B / \text{Eff}')$  / nil }
→ {tlvs :  $\emptyset$  ;  $\emptyset$  ;  $\emptyset \vdash \text{ttv } lr \S e$  }
--reset
→ {te :  $\Delta$  ;  $\emptyset \vdash e \S A'$  / (ttv lr :: E') }
→ {ten :  $\Delta$  ;  $\emptyset$  ,  $A' \vdash en \S T / \text{Eff}$  }
→ {tlr :  $\Delta$  ;  $\emptyset \vdash e' \S (L \text{ ttv } lr \text{ at } T / \text{Eff})$  / nil }
→ {tlvr :  $\Delta$  ;  $\emptyset \vdash \text{ttv } lr \S e$  }
→ (proj1 (mplug f (shift0 e' es) ts))  $\equiv$  e
→ reset0 e en e' , s
→ REExprSubstTyped._[_] es
  (lam (reset0 (proj1 (mplug (↑m {Γ' =  $\emptyset$  , T} f) (var 0) (λ-var Z))) en e'))
  {te = tes} {te1 = gvalue {E = Eff} vlam cont-type}
  .proj1 , s
-}

```

Since simple reduction above is defined directly on redexes, we introduce $\rightarrow$ that represents reduction within metaframe. As we are only considering whole typed expressions, in place of Γ we use empty context.

```

{-
infix 2 _→_
data _→_ : REExpr × State → REExpr × State → Set where
  →frame :  $\forall \{e1 \ e1' \ e2 \ e2' \ s \ s' \ n \ \Delta \ \Delta' \ A \ T \ \text{Eff} \ \text{Eff}' \ t1 \ t2\} \rightarrow (f : \text{Metaframe } \Delta \ \emptyset \ T \ \text{Eff} \ A \ \text{Eff}' \ \Delta' \ n)$ 
    → e1' , s → e2' , s'
    → Data.Product.proj1 (mplug f e1' t1)  $\equiv$  e1
    → Data.Product.proj1 (mplug f e2' t2)  $\equiv$  e2
    → (e1 , s) → (e2 , s')
-}

```

4.3 Progress

```

{-
data Decompose : State → (e : REExpr) → Set where
  de-simpl-redex :  $\forall \{e \ e2 \ s \ s' \ n \ \Delta \ \Delta' \ A \ T \ \text{Eff} \ \text{Eff}'\}$ 
    → (f : Metaframe  $\Delta \ \emptyset \ T \ \text{Eff} \ A \ \text{Eff}' \ \Delta' \ n$ )
    → (e , s) → (e2 , s')
    → (t :  $\Delta$  ;  $\emptyset \vdash e \S A / \text{Eff}$ )
    → Decompose s e
  de-shift :  $\forall \{s \ \Delta \ \Delta' \ T \ \text{Eff} \ A \ n \ \text{Eff}' \ es \ es' \ e \ l \ t\}$ 
    → (f : Metaframe  $\Delta \ \emptyset \ T \ \text{Eff} \ A \ \text{Eff}' \ \Delta' \ n$ )
    → shift0 (label l) es'  $\equiv$  es
    → Data.Product.proj1 (mplug f es t)  $\equiv$  e
    → (t :  $\Delta$  ;  $\emptyset \vdash e \S A / \text{Eff}$ )
    → (ts :  $\Delta$  ;  $\emptyset \vdash es \S T / \text{Eff}'$ )
    → Decompose s e
  de-val :  $\forall \{\Delta \ \text{Eff} \ A \ s \ e\} \rightarrow \{t : \Delta ; \emptyset \vdash e \S A / \text{Eff}\}$ 
    → Value e
    → Decompose s e
data Progress : State → REExpr → Set where
  done :  $\forall \{e \ s\} \rightarrow \text{Value } e \rightarrow \text{Progress } s \ e$ 
  step :  $\forall \{e1 \ s1 \ e2 \ s2\}$ 
    → (e1 , s1) → (e2 , s2)
    → Progress s1 e1
-}

```

Proof of progress would have a type of $\text{progress} : \forall \{A \ \Delta \ \text{Eff}s\} \rightarrow (s : \text{State}) \rightarrow (e : \text{REExpr}) \rightarrow (t : \Delta ; \emptyset \vdash e \S A / \text{Eff}s) \rightarrow \text{Progress } s \ e$. In such proof We would use auxiliary struct **Decompose** which builder would walk down well typed expression recursively until it has reached either value, simple reduction (app, tapp, new), or shift, and return it with the surrounding metaframe.

In case of shift, such a metaframe by construction should have an effect handler that has the same effect as shift. So we can construct **reset₀-k** and surrounding metaframe. Other cases would either be immediate value, or simple reduction in context.

4.3.1 Preservation

Current definition of reduction relation would yield proof of preservation immediately if progress was given since, frames and expressions are well typed, and reduction relation requires proofs to plug into metaframes or substitution.

Chapter 5

Discussion/Closing thoughts

In this paper we presented a language with labeled delimited effect handlers together with partial formalisation of such language. We defined 2 term languages, typing judgements for them, translations between those languages and that typing is preserved. We defined intrinsically typed frames and metaframes that allow us to define reduction relation, how it preserves types. Since they are typed we statically know what effects are handled by frame. This formalisation could be expanded and finished to fully prove correctness of the language and translations.

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